ASS ASSIGNMENT-8.5

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BATCH-45

Task Description #1 (Username Validator – Apply AI in

Authentication Context)

• Task: Use AI to generate at least 3 assert test cases for a

function is_valid_username(username) and then implement

the function using Test-Driven Development principles.

• Requirements:

o Username length must be between

5 and 15 characters. o Must contain only

alphabets and digits. o Must not start with

a digit.

o No spaces allowed. Example Assert

Test Cases:

assert is_valid_username("User123") == True

assert is_valid_username("12User") == False assert

is_valid_username("Us er") == False

Expected Output #1:

- Username validation logic successfully passing all AI- generated test cases.

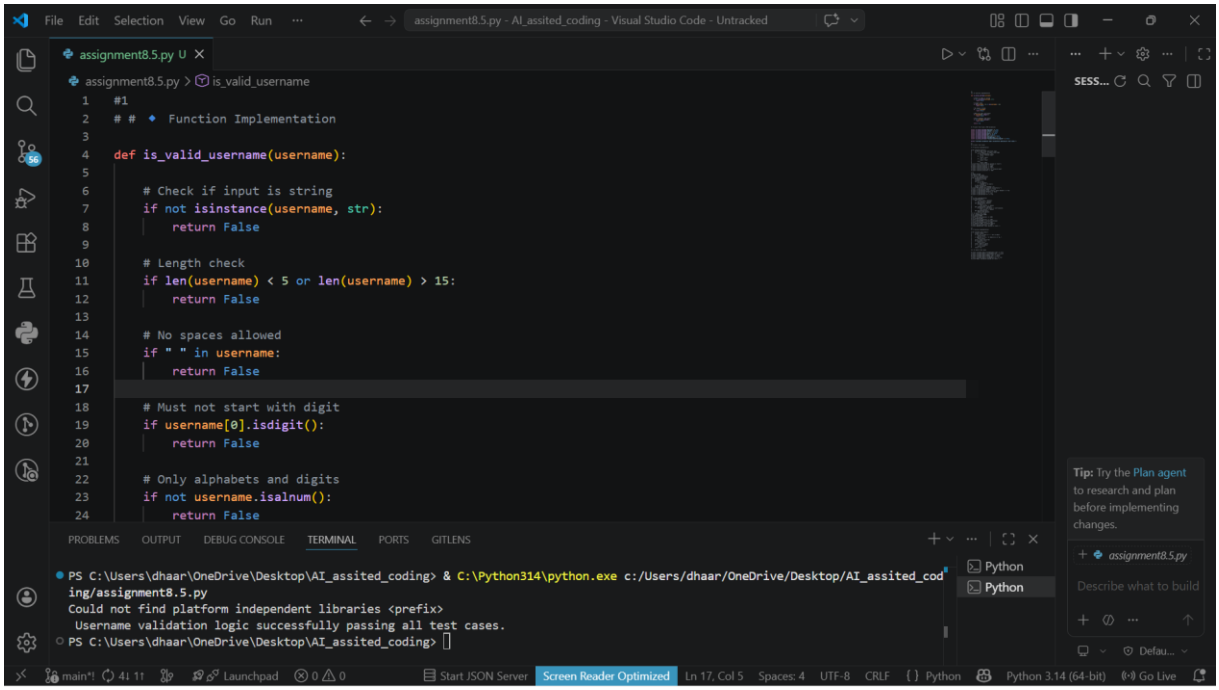
PROMPT:

Generate at least 3 assert-based test cases for a function `is_valid_username(username)`.
Requirements:

- Length between 5 and 15 characters
- Only alphabets and digits
- Must not start with a digit
- No spaces allowed

Then implement the function using Test-Driven Development principles.

OUTPUT:



Task Description #2 (Even–Odd & Type Classification – Apply

AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function classify_value(x) and implement it using conditional logic and loops.
- Requirements:

o If input is an integer, classify as "Even" or "Odd".

o If input is 0, return "Zero".

o If input is non-numeric, return "Invalid Input".

Example Assert Test Cases:

assert classify_value(8) == "Even" assert

classify_value(7) == "Odd" assert

classify_value("abc") == "Invalid Input"

Expected Output #2:

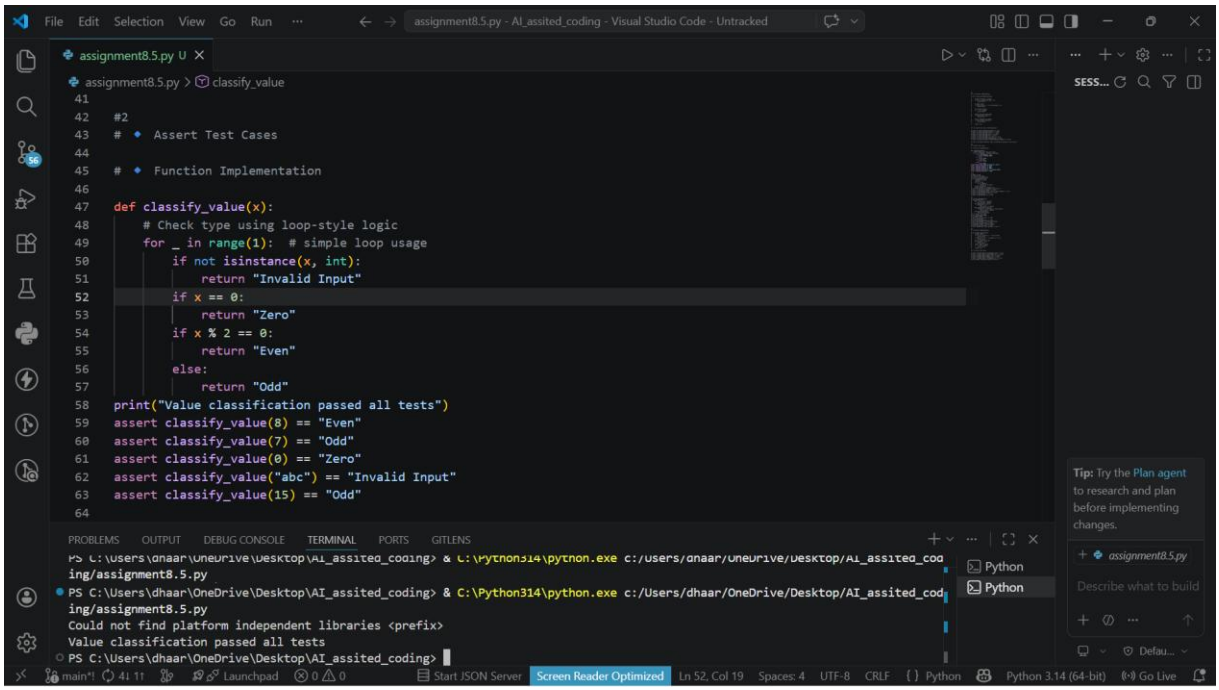
- Function correctly classifying values and passing all test cases.

PROMPT:

Generate at least 3 assert-based test cases for a function `classify_value(x)`.
Requirements:

- If integer → classify as "Even" or "Odd"
- If 0 → return "Zero"
- If non-numeric → return "Invalid Input"
  Implement using conditional logic and loops.

OUTPUT:



Task Description #3 (Palindrome Checker – Apply AI for

String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function

  is_palindrome(text) and implement the function.

- Requirements:

o Ignore case, spaces, and punctuation.

o Handle edge cases such as empty strings and single

characters.

Example Assert Test Cases:

assert is_palindrome("Madam") == True assert

is_palindrome("A man a plan a canal Panama") == True

assert is_palindrome("Python") == False

Expected Output #3:

- Function correctly identifying palindromes and passing all

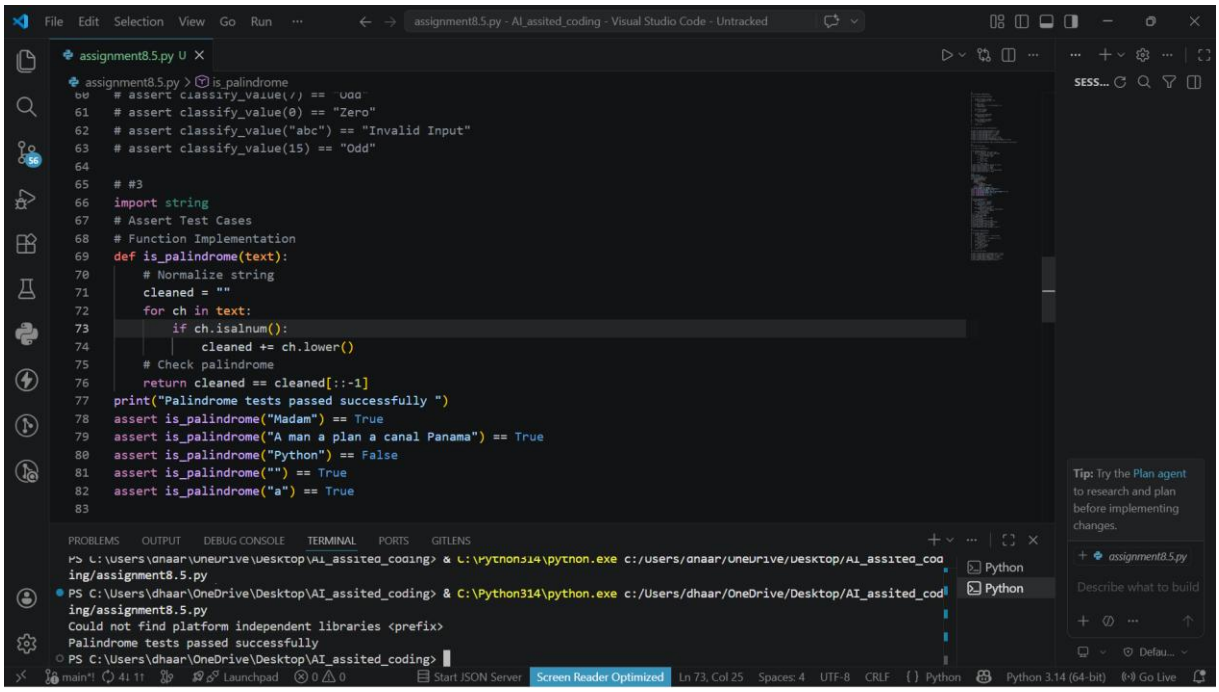
AI-generated tests.

PROMPT:

Generate at least 3 assert-based test cases for a function `is_palindrome(text)`.
Requirements:

- Ignore case, spaces, punctuation
- Handle empty string and single character Implement the function accordingly.

OUTPUT:



Task Description #4 (BankAccount Class – Apply AI for

Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases

  for a BankAccount class and then implement the class.

- Methods:

o deposit(amount) o

withdraw(amount) o

get_balance()

Example Assert Test Cases: acc =

BankAccount(1000)

acc.deposit(500) assert

acc.get_balance() == 1500

acc.withdraw(300) assert

acc.get_balance() == 1200

Expected Output #4:

- Fully functional class that passes all AI-generated assertions.

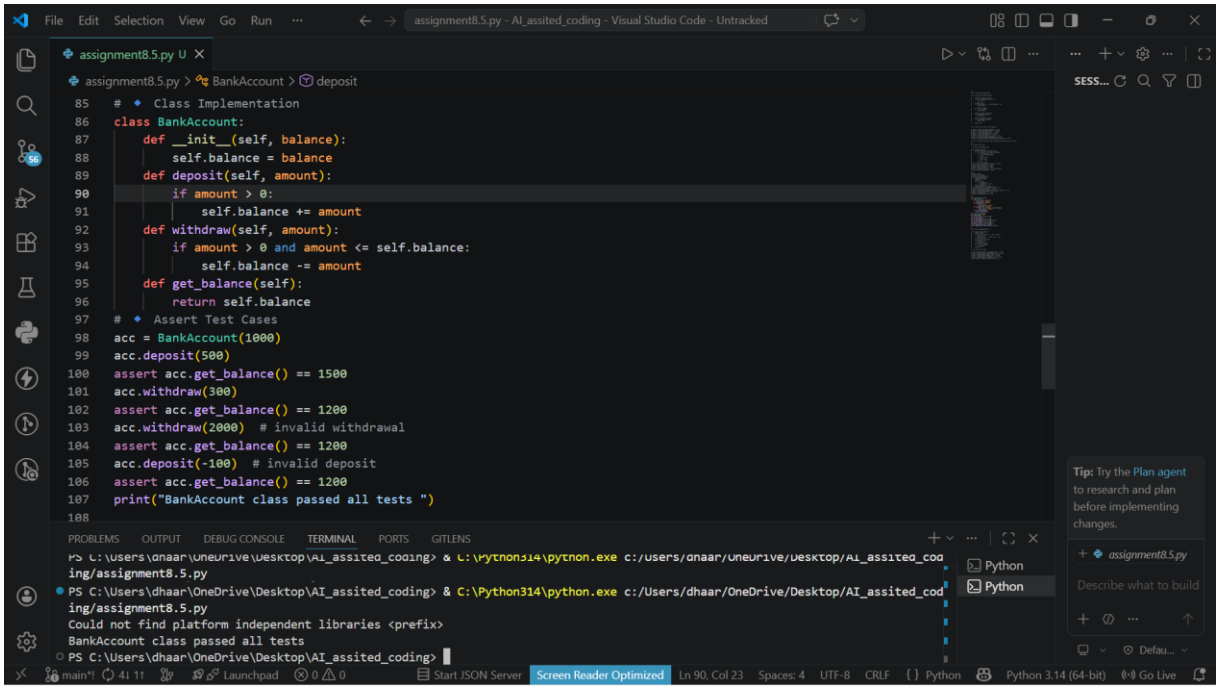
PROMPT:

Generate at least 3 assert-based test cases for a `BankAccount` class. Methods:

- deposit(amount)
- withdraw(amount) • get_balance()

Implement the class ensuring it passes all tests.

OUTPUT:



Task Description #5 (Email ID Validation – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for a function validate_email(email) and implement the function.

- Requirements:

o Must contain @ and .

o Must not start or end with special characters. o

Should handle invalid formats gracefully.

Example Assert Test Cases:

assert validate_email("user@example.com") == True

assert validate_email("userexample.com") == False assert

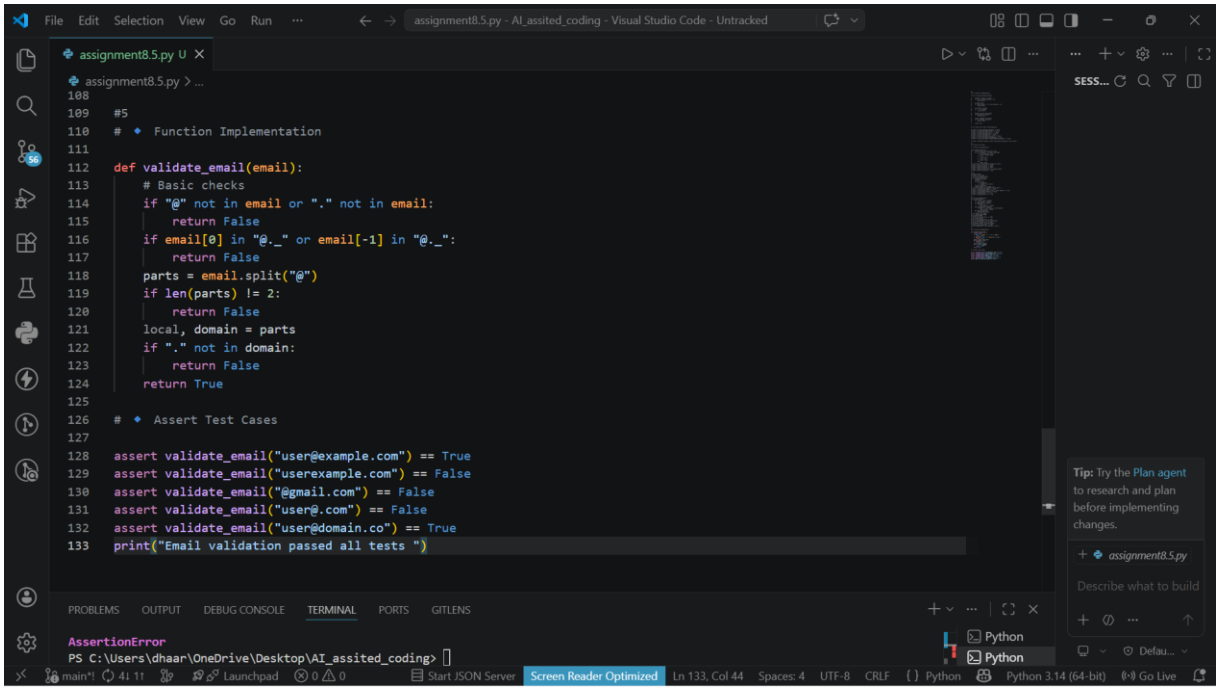
validate_email("@gmail.com") == False

Expected Output #5:

- Email validation function passing all AI-generated test cases and handling edge cases correctly.

PROMPT:

Generate at least 3 assert-based test cases for a function `validate_email(email)`.
Requirements:

- Must contain @ and .
- Must not start or end with special characters
- Handle invalid formats gracefully
  Implement the function accordingly.

OUTPUT: